



Convert LaTeX to HTML

Is there a converter that will convert LaTeX to HTML? I have tried several services but the LaTeX diagrams and some of the maths came out weird.

I actually converted the Latex to PDF without a problem. Then I used [an online PDF to HTML converter](#) to convert from PDF to HTML. The square root signs did not appear properly and some of the Cartesian coordinate plane diagrams had nothing on them except the grid.

```
\documentclass[12pt,a4paper]{report}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage{amsmath}
\usepackage{pstricks,pst-plot}
\usepackage{pst-eucl}
\addtolength{\hoffset}{0.2cm}
\addtolength{\textwidth}{1.1cm}
\addtolength{\textheight}{1.0cm}
\addtolength{\voffset}{-0.7cm}
\linespread{1.3}
\begin{document}

If  $x$  is non-positive then  $x \leq -\sqrt{\frac{1}{2}}$   $\hspace{30 mm}$   $(6)$ 

(4) and (6) tell us that

(4)  $-1 \leq x \leq 0$ 

(6) If  $x$  is non-positive then  $x \leq -\sqrt{\frac{1}{2}}$ 

Therefore  $-1 \leq x \leq -\sqrt{\frac{1}{2}}$ 

\hspace{50 mm}

\begin{align*}
\psgrid[unit=2cm](0,0)(-2,-2)(2,2)
\pnode(-1.8,.87){b}\uput[50](b){b}
\pnode(-1.8,1.8){a}\uput[50](a){a}
\psset{linecolor=green}
\psline[linewidth=1pt](-1.8, 0)(-1.8,4)
\psset{linecolor=green}
\psline[linewidth=1pt](-1.8, 0)(-1.8,-4)
\psset{linecolor=blue}
\psellipticarc(0,0)(2,2){0}{180}
\psset{linecolor=red}
\psline[linewidth=1pt,linestyle=dashed,dash=2pt](-2,2)(-1.41,1.41)
\psset{linecolor=red}
\psline[linewidth=1pt](-1.41,1.41)
\psset{linecolor=red}
\psline[linewidth=1pt]{->}(4,-4)
\psset{linecolor=red}
\psline[linewidth=1pt]{->}(-2,2)(-4,4)
\end{align*}

\end{document}
```

PDF output:

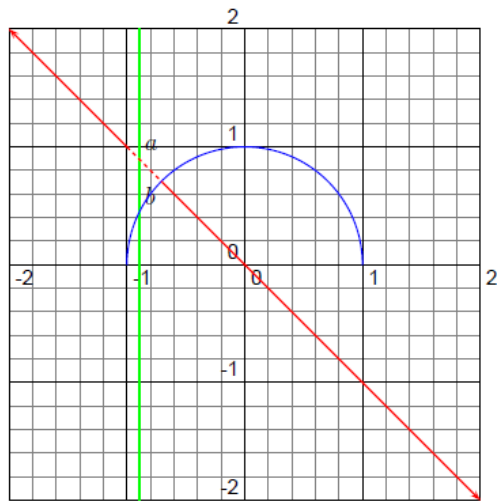
If x is non-positive then $x \leq -\sqrt{\frac{1}{2}}$ (6)

(4) and (6) tell us that

$$(4) -1 \leq x \leq 0$$

(6) If x is non-positive then $x \leq -\sqrt{\frac{1}{2}}$

Therefore $-1 \leq x \leq -\sqrt{\frac{1}{2}}$



PDF->HTML output:

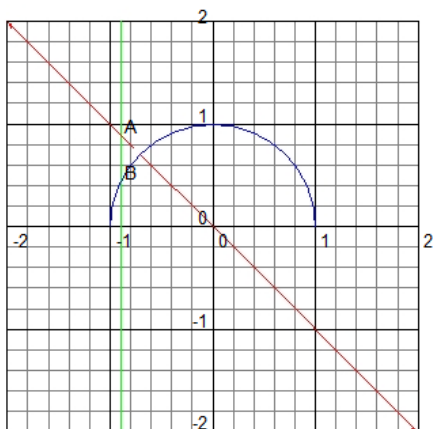
If x is non-positive then $x = -\frac{1}{2}(6)^{\frac{1}{2}}$

(4) and (6) tell us that

(4) $-1 = x = 0$

(6) If x is non-positive then $x = -\frac{1}{2}$

Therefore $-1 = x = -\frac{1}{2}$



{pdftex} {conversion} {html} {latex2html}

edited Jun 14 at 22:56



David Carlisle
399k 33 948
1640

asked Dec 25 '11 at 14:52



XYZ
494 2 7 8

2 It may help if you describe what techniques you have tried and include examples of what turned out weird. – **N.N.** Dec 25 '11 at 14:57

4 Perhaps one of the questions listed at tex.stackexchange.com/questions/tagged/latex2html is helpful to you? – **doncherry** Dec 25 '11 at 15:18

1 @doncherry: And if those don't, there's always the [FAQ](#). – **Martin Schröder** Dec 25 '11 at 18:19

1 So, @XYZ: [Welcome to TeX.sx!](#) Please add a [minimal working example \(MWE\)](#) that illustrates your problem. – **doncherry** Dec 25 '11 at 19:08

See comment below answer. – **XYZ** Dec 26 '11 at 13:52

9 Answers

Using PDF as an intermediate format when converting from LaTeX to HTML is not very good idea. LaTeX and HTML are both mostly structural markup languages, which means you use them to describe the document structure (sections, emphasize, formulas etc.), whereas PDF is mostly about representation of your document on the screen or paper. When converting LaTeX to PDF, you lose much of the structural information, and it cannot be successfully recovered by conversion from PDF to HTML.

It is much better to convert LaTeX directly to HTML. There are [number of ways](#) (WayBack Archive) how to do that, one I would recommend is by using `htlatex`. It is probably already part of your TeX distribution, is very powerful and flexible, and its use can be as simple as running

```
htlatex mydocument.tex
```

If you tell us more about your environment (which operating system do you use, what is your TeX distribution, your text editor/LaTeX IDE, how you generated the PDF file etc.) we may be able to give you more details on how to use `htlatex` .

edited Jul 16 '13 at 2:58

answered Dec 27 '11 at 0:47



Jan Hlavacek

14.5k 3 43 71

I used MiKTeX 2.9 which I downloaded off the internet onto a computer using Windows 7. It is TeXworks and some of the options are pdfLaTeX + MakeIndex + BibTeX, LaTeXmk, pdfTeX, pdfLaTeX, pdfLaTeX + MakeIndex + BibTeX, XeTeX, XeLaTeX, XeLaTeX + MakeIndex, + BibTeX, ConTeXt(LuaTeX), conTeXt(pdfTeX), ConTeXt(XeTeX) - XYZ Dec 27 '11 at 14:04

To get htlatex on Ubuntu 14.04: `sudo apt-get install tex4ht` - kebs Apr 5 at 13:24

Is there a converter that will convert LaTeX to HTML?

Yes. **There are** lots. But you already knew this, as evidenced by your second "question":

I have tried several services but the LaTeX diagrams and some of the maths came out weird.

"turning out weird" could mean any number of things:

- The document fails to compile
- The document compiles, but is empty or corrupted
- The document compiles and opens, but is somehow faulty.

Each of these categories breaks down further and the solution may well depend on what you used to do the conversion, so as it stands, this question is unanswerable.

For diagrams, one word of general advice I would give is that it might be best to compile the diagrams separately and include them into the html by hand with `<img>` tags. Diagrams are hard. The `standalone` package/class can help with this. See also the [pgf/TikZ externalisation tag](#).

edited Apr 13 at 12:34

answered Dec 25 '11 at 21:27



Community ♦

1



Seamus

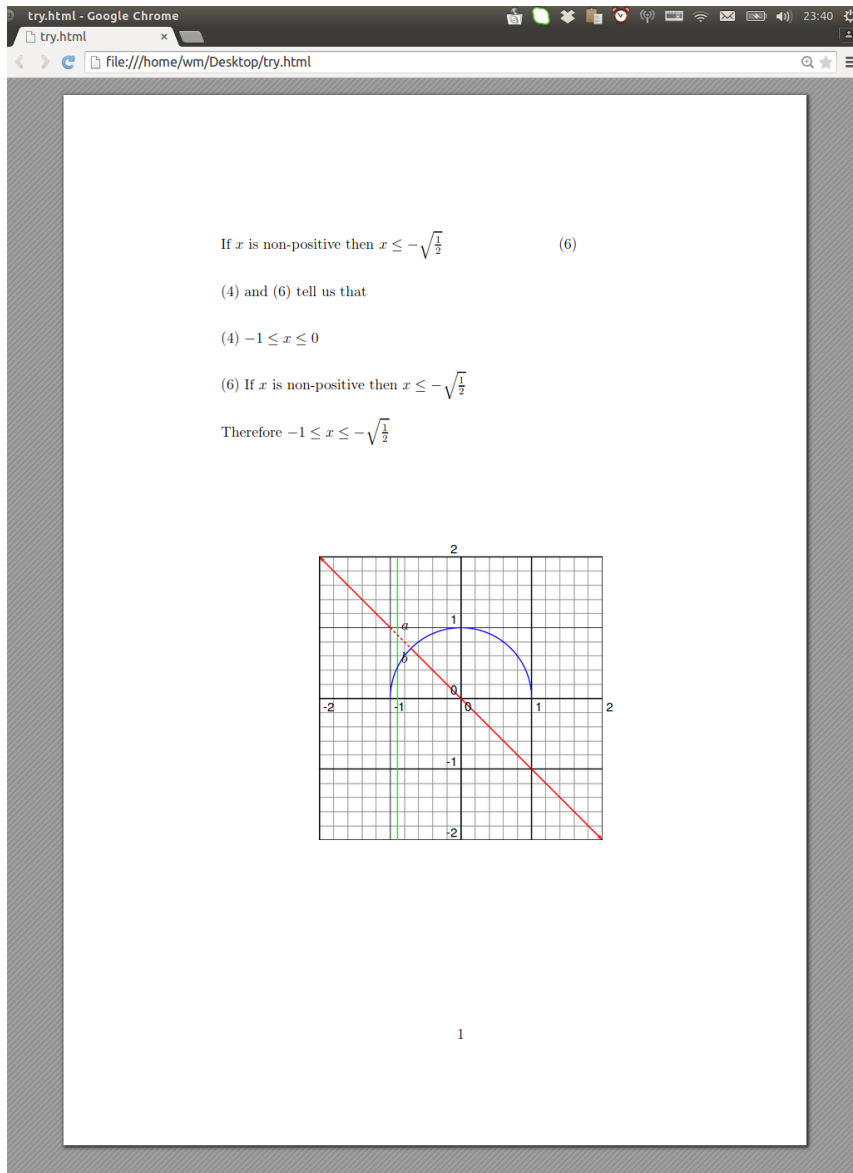
39.8k 32 177 309

I actually converted the Latex to PDF without a problem. Then I used www.pdfonline.com to convert from PDF to HTML. The square root signs did not appear properly and some of the Cartesian coordinate plane diagrams had nothing on them except the grid. - XYZ Dec 26 '11 at 13:48

3 @XYZ add this information to your question, along with an example of a file that has this problem. - Seamus Dec 26 '11 at 17:13

You may try to use [pdf2htmlEX](#), which is a general PDF to HTML converter. You can compile the LaTeX source to PDF, and then convert the PDF to HTML via pdf2htmlEX. A good reference of pdf2htmlEX is its [wiki page](#). The work-flows are given in the Figure 3 of the article on TUGboat: [Online publishing via pdf2htmlEX HTML](#) / [PDF](#).

The screenshot shows the converted HTML (Ubuntu 14.04, Google Chrome 43).



edited Jun 3 '15 at 15:45

answered May 25 '15 at 16:42



Wanmin Liu

101 1 2

Wow, pdf2htmlEX looks awesome: coolwanglu.github.io/pdf2htmlEX/demo/cheat.html - asmaier Sep 20 '15 at 20:44

This tool worked perfectly for me, and seems a lot more trustworthy than many other tools displayed by google. - mafu Sep 16 '16 at 12:05

I have had good mileage with and would recommend **pandoc** for converting LaTeX to HTML output.

Pandoc provides capability for applying CSS styling to HTML output and can take care of equation rendering with e.g. MathML and citation/bibliography styling, among many other features, and is well documented.

answered Jun 23 '15 at 15:49



Harry

313 2 8

As previously observed using PDF format for converting from LaTeX to HTML is very bad idea. Here you will find a comprehensive list of programs that convert LaTeX to HTML (Note not all of those programs are capable of converting pure TeX.)

<http://www.tex.ac.uk/cgi-bin/texfaq2html?label=LaTeX2HTML>

I am familiar with three of these programs: LaTeX2HTML, TeX4ht, and Hevea. All of them have their strengths and weaknesses and they use different approaches. TeX4ht is included in TeXLive and MiKTeX unlike other two and from the point of view of theoretical CS is a "correct" way to do conversion (not surprisingly it was written by a (now deceased) member of the CS department at Ohio State). If you have just a few pages to convert, probably the right approach for you would be to choose any of the three and then do some manual editing. Now you have to be careful. For example Hevea will try to use HTML tags for things like square root while TeX4ht will use bitmap images for any math expressions. Since pure HTML has a poor support for mathematics the first approach can quickly get into trouble but when it works the output is really beautiful.

As for diagrams and pictures, all of the above programs to my knowledge rely on ImageMagic to do image conversion. Hevea has to be run multiple times (it is actually bunch of scripts) and the proper way to use it is by writing a Makefile (Unix again). I would actually convert images and diagrams manually using GraphicMagick which is IMHO far more stable and better batch image processor than ImageMagic.

edited Dec 27 '11 at 17:25



Alan Munn

131k 22 367 586

answered Dec 27 '11 at 5:51



Predrag Punosevac

6,274 1 37 59

What do you mean by "theoretical CS correct" more precisely? – [Ciro Santilli 709大抓捕 六四事件 法轮功](#)
Aug 15 '14 at 14:54

Three words: use LaTeX subset

LaTeX and its common output formats like PDF are super complicated. Converting all of it to HTML + CSS is close to impossible.

But if you restrict yourself only to a small subset of LaTeX features it becomes feasible.

Check out [Softcover](#): it has a LaTeX subset which it calls Polytex that it can convert to HTML. The exact subset is not well documented, but it already handles 99.9% of what we need to make regular books as shown in their documentation https://github.com/softcover/softcover_book/tree/master/chapters : sections, lists, equations, tables, images, references. Softcover does its magic through [Tralics](#), which does LaTeX to XML.

As Seamus said, diagrams are hard, and it is unlikely they will be part of the implementable subset any time soon, so use images. Generate the images through a command line tool like gplot or pyplot, and then build the PDF with a Makefile that first builds the images and then the PDF.

answered Aug 15 '14 at 14:38



[Cyrus Dabizadeh](#) 709大抓捕 六四事件 法轮功

226 1 10

If anyone can guess why the downvote, let me know. – [Cyrus Dabizadeh](#) 709大抓捕 六四事件 法轮功 Jul 11 '15 at 14:36

[RtextDoc](#) has a built in Latex to Html converter

answered Oct 23 '13 at 4:57



frey

21 1

2 How well does it cope with the type of documents the poster is interested in? – [Andrew Swann](#) Oct 23 '13 at 6:25

I'm currently developing a free open source tool that can convert LaTeX to a single HTML file. Its called PDBF (<https://github.com/uds-datalab/PDBF>) and runs on win/linux/mac.

Instructions:

- Make sure you have a Java Runtime (version ≥ 1.7) installed

- Download latest version of PDBF: <https://github.com/uds-datalab/PDBF/archive/gh-pages.zip>
- Unpack archive
- Change current dir to unpacked folder
- Run: `java -jar pdbf.jar /path/to/sometexfile.tex`

Resulting HTML file is placed in the same folder as the source tex file with the same name as the source tex file.

Technical details: The PDBF compiler basically uses a regular tex engine to compile your pdf and then stores the pdf as base64 encoded javascript string in the HTML and then includes a slightly modified version of pdf.js (The free and open source pdf engine of firefox) into the HTML to display that pdf.

Hope you like my project. Feel free to contact me through github if you have suggestions or encounter bugs.

edited Sep 29 '15 at 19:18

answered Sep 29 '15 at 19:12



Patrick Bender

58 7

What do you gain from running the standard PDF previewer of Chrome/Firefox, etc ...? - Paulo Ney Feb 15 at 2:59

@Paulo Ney 1. You can then open the html files also with browsers that have no PDF viewer included. 2. It allowed us to extend PDF.js to make charts and other things in the resulting html interactive ([example](#)) - Patrick Bender Feb 16 at 6:35

You might want to consider using [DocOnce](#) as primary format instead of LaTeX:

DocOnce is a modestly tagged (Markdown-like) markup language targeting scientific reports, software documentation, books, blog posts, and slides involving much math and code in the text. From DocOnce source you can generate LaTeX, Sphinx, HTML, IPython notebooks, Markdown, MediaWiki, and other formats. This means that you from a single source can get the most up-to-date publishing technologies for paper, tablets, and phones.

answered Oct 22 '16 at 17:28



asmaier

956 1 6 6